

EXÁMENES FINALES PRIMER PERIODO

GRADO: 6

CIENCIAS NATURALES

1. sdsdsdsd

A.

KALETH MAZA

B.

KALETH MAZA

C.

KALETH MAZA

D.

KALETH MAZA

2. ddddd

[illegible]

```

graph TD
    Start([Start]) --> Feasible{Is the problem feasible?}
    Feasible -- No --> End([End])
    Feasible -- Yes --> InitPop[Generate initial population]
    Feasible -- Yes --> InitSol[Generate initial solutions]
    Feasible -- Yes --> CalcFit[Calculate fitness]
    InitSol --> InitPop
    CalcFit --> TermCond{Is the termination condition satisfied?}
    TermCond -- No --> UpdatePop[Update the population]
    UpdatePop --> NewSol[Generate new solutions]
    NewSol --> CalcFitNew[Calculate fitness]
    CalcFitNew --> TermCond
    TermCond -- Yes --> OutputSol[Output the optimal solution]
  
```

```

classDiagram
    class CirugiaController {
        +load_cirugias() : DataFrame
        +save_cirugias()
        +validate_preop(data) : tuple
        +validate_intraop(data) : list
        +validate_postop(data) : list
        +export_preop, postop, diag : str
        +render_cirugia()
    }
    class HistoricaController {
        +load_historia() : DataFrame
        +save_historia()
        +checklist_completo(row) : bool
        +render_historia_cinica()
    }
    class Auth {
        +cargar_usuario() : DataFrame
        +login()
    }
    class CSVDataStore {
        +cirugias.csv
        +historia_clinica.csv
        +pacientes.csv
        +usuarios.csv
    }
    class ClinicalAssistant {
        +client : Gnuq
        +available_models : dict
        +init(not_key : str)
        +prepare_clinical_context(paciente, cirugias, historia) : str
        +ask_question(context, question, model) : dict
        +summarize_clinical_record(context) : dict
        +get_api_key() : str
        +parse_json_safe(value) : dict
    }
    CirugiaController --> CSVDataStore : use/subscribe
    HistoricaController --> CSVDataStore : use/subscribe
    Auth --> ClinicalAssistant : use
    
```

[illegible]

```

graph TD
    Start([Start]) --> Init[Initialize problem]
    Init --> IsProblem{Is there any problem?}
    IsProblem -- No --> Output1[Output: Pareto frontier]
    IsProblem -- Yes --> GenSol[Generate random solutions]
    GenSol --> EvalFit1[Evaluate fitness]
    EvalFit1 --> IsImproved1{Is the fitness improved?}
    IsImproved1 -- No --> GenNewSol[Generate new solutions]
    IsImproved1 -- Yes --> UpdatePF1[Update Pareto frontier]
    GenNewSol --> EvalFit2[Evaluate fitness]
    EvalFit2 --> IsImproved2{Is the fitness improved?}
    IsImproved2 -- No --> GenNewSol
    IsImproved2 -- Yes --> UpdatePF2[Update Pareto frontier]
    UpdatePF1 --> Output2[Output: Pareto frontier]
    UpdatePF2 --> Output2
    
```

CIENCIAS NATURALES

3. dsfsdf

- A. dsfsd
- B. sdfsdf
- C. dsfsd
- D. sdfsdf

4. dsfsdf

- A. dsfsdf
- B. sdfsdf
- C. sdfsdf
- D. sdf

5. sdfsdf

- A. dsfsdf
- B. dsfsdf
- C. sdfsdf
- D. dsfsdf

6. dsfsdf

- A. ewf
- B. sdfsdf
- C. dsfsdf
- D. sdfsdf

7. fsdfsdf

- A. dsfsdf
- B. sdfsdf
- C. dsfsdf
- D. sdfsdfsdf

8. fdsfsdfsdf

- A. sdfsdf
- B. dsfsdf
- C. dsfsdfsdf
- D. dsfsdf

CIENCIAS NATURALES

9. dsfsdf

- A. dsfsdf
- B. sdfdsf
- C. dsfsdf
- D. fsdf

10. dsfsdfsdf

- A. dsfsdf
- B. sdfdsf
- C. dsfsdf
- D. dsfsdf

11. sdfsdffsdf

- A. sdfsdffsdf
- B. dsfdsf
- C. dsfdsf
- D. sdfdsf

12. dsfsdfsdf

- A. dsfsdf
- B. fsdfsd
- C. fdsfsd
- D. dsfsdf

fsd

13. dsfsdfsdfsdf

- A. dsfsdf
- B. dsfsdf
- C. dfsdf
- D. dsfsdf

CIENCIAS NATURALES

14. sdfsd sdf

- A. sdfsd
- B. sdfsd
- C. sdfsd
- D. sdfsd

15. dsf dsf

- A. sdfsd sdf
- B. sdfsd sdf
- C. sdfsd
- D. dsf

16. sdfsd sdf

- A. sdfsd sdf
- B. dsf
- C. sdfsd
- D. fsdf

17. dsf dsf

- A. sdfsd sdf
- B. fsdf
- C. sdfsd
- D. sdfsd

18. sdfsd sdf

- A. fsdfsd
- B. sdfsd
- C. dsf dsf
- D. sdfsd

19. fsdfsd

- A. dsf dsf
- B. sdfsd
- C. dsf dsf
- D. sdfsd

CIENCIAS NATURALES

20. dfsfsdfsdfs

- A. fsdfdsfsd
- B. sdfsd
- C. sdfsd
- D. sdfsd